Hardware Library Tutorial

Dropping standard ISO metric (M3–M12) and UNC/UNF imperial (#4-40 to 1/2-13) fasteners into your model, with the right clearance bores and counterbores.

ForgeSlicer's Hardware Library ships **15 fastener grades** (7 metric + 8 imperial) covering everything from #4-40 chassis screws to 1/2-13 structural bolts. Each grade is a single click to drop a full bolt + clearance bore + counterbore + nut assembly.

1. What's in the library?

Every entry is a **real engineering spec** — the major thread radius, pitch, head height, and nut height come from ISO 4014 (hex bolts) / ISO 4762 (cap screws) for metric, and ANSI/ASME B18.6.3 for imperial. When you drop an M5×30, what shows up in the scene is geometry that matches the real part to a tenth of a millimeter.

ISO metric grades

Grade	Diameter	Coarse pitch	Head width (across flats)	Head height	Nut height
M3	3.0 mm	0.5 mm	5.5 mm	2.0 mm	2.4 mm
M4	4.0 mm	0.7 mm	7.0 mm	2.8 mm	3.2 mm
M5	5.0 mm	0.8 mm	8.0 mm	3.5 mm	4.0 mm
M6	6.0 mm	1.0 mm	10.0 mm	4.0 mm	4.8 mm
M8	8.0 mm	1.25 mm	13.0 mm	5.3 mm	6.5 mm
M10	10.0 mm	1.5 mm	16.0 mm	6.4 mm	8.0 mm
M12	12.0 mm	1.75 mm	19.0 mm	7.5 mm	10.0 mm

Available lengths per grade are filtered automatically — the picker won't let you build an M3×80 (silly) or an M12×6 (impossible to thread).

UNC / UNF imperial grades

Grade	Major dia.	TPI	Pitch (mm)	Head width	Nut height
#4-40	0.112"	40	0.635	5.30 mm	2.5 mm
#6-32	0.138"	32	0.794	6.40 mm	2.7 mm
#8-32	0.164"	32	0.794	7.60 mm	3.2 mm
#10-24	0.190"	24	1.058	8.80 mm	3.6 mm
1/4-20	0.250"	20	1.270	11.10 mm	5.6 mm
5/16-18	0.3125"	18	1.411	13.90 mm	6.5 mm
3/8-16	0.375"	16	1.588	16.70 mm	7.5 mm
1/2-13	0.500"	13	1.954	22.30 mm	10.0 mm

All imperial dimensions are **stored in millimetres internally**. The picker labels them in inches for familiarity, but the geometry is the same metric engine as the M-series — you can mix metric and imperial fasteners in the same assembly without unit headaches.

2. Dropping a fastener

Open the workspace, click **Library** → **Hardware** in the left toolbar (or use **Shift+H**). The dialog shows two tabs — **Metric** and **Imperial**. Inside each tab:

- **Grade** picker — pick M3, M5, 1/4-20, etc.

- **Length** picker — auto-filtered to commonly-available lengths for that grade.
- **Work thickness** (optional) — the host plate's thickness. Defaults to *length – 5 mm* so 5 mm of shaft pokes past the nut for tightening.
- Hit **Add to scene**. Four objects appear, all grouped under a single assembly name (e.g. *Fastener M5×30*): the bolt, a clearance bore, a head counterbore, and a nut.

3. Anatomy of a Fastener Pair

Each fastener assembly contains four members. Three are POSITIVE (geometry added to the scene) and one — wait, none are. Let me tell you what's actually going on:

Member	Modifier	Why it's there
Bolt	Positive	The visible bolt — a threaded shaft + hex head, generated from the grade's pitch and head dims. Goes through the assembly.
Bolt bore	Negative	A cylindrical hole 0.4 mm larger than the bolt's outer radius. Subtracted from any host the assembly is placed against, so the bolt fits cleanly.
Head counterbore	Negative	A wider recess (headR + 0.5 mm) carved at the top of the host. Lets the bolt head sit flush with the work surface, no protrusion.
Nut	Positive	A real hex nut at the bottom, sized to match the grade. Visual + functional — if you're 3D printing the assembly itself, the nut prints as a separate part.

The three negative members are **only** subtracted from primitives they overlap with. If you drop a fastener in empty space, you just see the bolt + nut. Move the assembly so the bolt-bore cylinder pierces your host plate, and the hole appears automatically — no manual cylinder placement.

4. Walkthrough — bolting two plates together

End-to-end recipe. Goal: two 60×60×6 mm aluminium-style plates joined by an M5×30 cap screw, drawn entirely in ForgeSlicer.

1. From the Primitives tab, drop a **Cube**. Set dims to 60×6×60 mm — this is plate A.
2. Drop another Cube with the same dims. Move it 12 mm up the Y axis (just above plate A) — this is plate B.
3. Open **Library** → **Hardware** (Shift+H). Pick **Metric** → **M5**, length **30 mm**, leave work-thickness on auto.
4. Hit **Add to scene**. A four-member assembly drops at the world origin: bolt, bore, counterbore, nut.
5. In the Inspector, set the assembly's position to **(30, 0, 30)** — the centre of plate A's footprint. The bolt now spears through both plates; the bore and counterbore carve a clean hole in each.
6. Slice it. The G-code preview should show two plates joined by a circular bolt-shaped piece of geometry, with a counterbore at the top of plate B and a nut at the bottom of plate A.

Pattern this for production parts. Drop the assembly once, use **Ctrl+D + Mirror X** (or right-click → Duplicate → Array) to populate corner holes. The bore + counterbore travel with every copy.

5. Related composites

The Hardware Library handles full bolt+nut assemblies. For fastener-related *features* on your host part without a real bolt, check the **Combo** tab in the Left Panel:

Composite	Use	Modifier
Slot	Adjustable / slide-fit hole for a bolt	Negative (default)
Countersink	Flat-head bolt recess (cone cut)	Negative
Hex pocket	Captive nut socket (square or hex)	Negative
Gusset	Corner reinforcement triangle next to a bolt	Positive

These compose with the Hardware Library naturally — drop a hex pocket where the nut would otherwise float, and you get a captive nut socket that holds the nut without a wrench during assembly. Common in 3D-printed enclosures.

6. Print-and-fit tips

- **Clearance assumes a 0.4 mm offset.** Most FDM printers produce holes slightly tighter than designed. For tight-fit applications (press-fit M5 into PETG), the bolt may need a tap to start. For sliding fits, increase the bore by 0.2–0.4 mm via the Inspector after dropping the assembly.
- **Counterbore depth = head height + 0.2 mm.** Designed so the bolt head sits 0.2 mm BELOW the surface — no protrusion, no scuffing. If you want a domed protrusion, set the assembly's Y position 1–2 mm lower so the head pokes up.
- **The bolt is full-thread.** ForgeSlicer's bolt primitive draws threads along the entire shaft. If you want a partial thread (smooth shank under the head), use the Inspector to switch headStyle from *hex* to *socket*; the shoulder appears automatically.
- **Nut may be unprintable as part of a print.** Real metal nuts thread cleaner than printed ones. Hide the nut layer before slicing if you intend to use real hardware — or use the **Hex pocket** composite to hold it captive.
- **Imperial and metric are interchangeable.** Pick whichever is closer to your design intent. If you tap a 1/4-20 thread into a 6 mm hole or vice versa, the world doesn't end — the geometry is consistent.

7. Troubleshooting

Symptom	Likely cause	Fix
Bolt is visible but the hole isn't appearing in the host	The bore cylinder doesn't overlap the host — assembly was placed in empty space.	Move the assembly so the bore pierces the host. The hole appears automatically on the next CSG pass.
Two assemblies dropped on top of each other	Duplicate clicked too fast.	Ctrl+Z, then use the Inspector to position the second one before duplicating again.
Bolt printed but threads look stripped	Thread height < 0.4 mm + 0.2 mm layer height = invisible.	Print at 0.12 mm layers, or use a 0.2 mm nozzle. Or use the bolt primitive without threads (Inspector → headStyle → *simple*).
Counterbore is too shallow on a thin plate	Counterbore depth = headH + 0.2 mm, exceeds your plate thickness.	Switch to a flat-head bolt + Countersink composite — the cone cut uses less depth than a counterbore.
Need a fastener size that's not in the table	Library is intentionally limited to common shop sizes.	Drop the closest grade, then edit individual member dimensions in the Inspector. Or open *hardwareLibrary.js* and add a row to HARDWARE_TABLE.

8. Quick reference

Keyboard shortcut: Shift+H opens Hardware Library.
Source: grade tables live in `frontend/src/lib/hardwareLibrary.js`. Convert a grade + length to fastener opts via `hardwareToFastenerOpts(spec, length)`.
Dialog: `frontend/src/components/dialogs/HardwareLibraryDialog.jsx`.
Builder: `buildFastenerPair` in `frontend/src/lib/composites.js`.
Test: `frontend/tests/hardware-library.mjs`.

Tutorial generated May 29, 2026 from ForgeSlicer source. Run the corresponding `scripts/build_*_tutorial.py` *to regenerate after future updates.*